Toolkit: Chemistry of the Earth

Practical Nine: Water – Chemical Weathering

Objectives:

- Familiarization with the way chemical reactions are written for aqueous (electrolyte) solutions
- Understand why natural waters can be acidic
- Examine chemical weathering reactions to understand how rocks at the Earth's surface break down through chemical mechanisms

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FEEDBACK

CHEMICAL WEATHERING

Chemical weathering is the process by which minerals at Earth's surface interact with water (particularly acidic water) and the gases in the atmosphere (CO_2 and O_2) to form mineral assemblages that are thermodynamically more stable under the ambient pressure and temperature (P-T) conditions. Acids are produced from biogenic activity (plant activity, bacterial metabolism) as well as acidic rainwater.

Chemical weathering can involve:

- dissolution reactions e.g. dissolution of halite, carbonates
- hydration reactions e.g. formation of gypsum

Common *products* of chemical weathering reactions are

- clay minerals e.g. kaolinite, smectite formed in hydrolysis reactions
- oxides, hydroxides e.g. hematite, goethite, gibbsite formed in oxidation reactions

A. Carbonic Acid

Carbonic acid is a natural acid that forms when CO_2 from the atmosphere dissolves in water. We will look at the reactions involved in carbonic acid formation.

CO_2 in the atmosphere dissolves in water to form carbonic acid:

$$CO_{2\,(g)} + H_2O_{\,(l)} = H_2CO_{3\,(aq)} \qquad \text{(Reaction 1)}$$

Carbonic acid is a weak acid. It dissociates in two steps:

$$H_2CO_{3\,(aq)} = H^+ + HCO_3^- \qquad \text{Bicarbonate: } HCO_3^- \qquad \text{(Reaction 2)}$$

$$HCO_3^- = H^+ + CO_3^{2-} \qquad \text{Carbonate: } CO_3^{2-} \qquad \text{(Reaction 3)}$$

1. Equilibrium (dissociation) constants, K for the above reactions.

For Reaction 1, the equilibrium constant K_1 is written:

$$K_1 = \frac{[H_2CO_3]_{(aq)}}{[CO_2]_{(g)}\,[H_2O]_{(l)}}$$

Write the equilibrium constant for Reactions 2 and 3: (all the species are aq)

$$K_2 = \frac{[H^+]_{(aq)}\,[HCO_3^-]_{(aq)}}{[H_2CO_3]_{(aq)}} \qquad K_3 = \frac{[H^+]_{(aq)}\,[CO_3^{2-}]_{(aq)}}{[HCO_3^-]_{(aq)}}$$

2. For a very dilute solution, $[H_2O] = 1$, so

$$K_1 = \frac{[H_2CO_3]_{(aq)}}{[CO_2]_{(g)}}$$

We can **calculate a value for the equilibrium constant K_1**, knowing the solubility of CO_2 in water: (See Example 4-1, Misra, p. 53)

a) At 1 bar and 298.15 K, the solubility of CO_2 in water is 0.148 g of CO_2 per 100 g of water. We need to convert this to the units mol kg^{-1}.

First convert to g kg^{-1}: 1.48 g kg^{-1}

Now convert to mol kg-1: (Atomic mass for C = 12, O = 16)

Molecular weight of CO_2 = 12 + (16*2) = 44

1.48 g = 1.48 / 44 = 0.0336 mol

Solubility = 0.0336 mol kg^{-1}.

b) We assume that every mole of CO_2 that dissolves produces one mole of $H_2CO_{3\,(aq)}$.

Therefore, the number of moles of $H_2CO_{3\,(aq)}$ in 1 kg of solution = number of moles CO_2 = 0.0336 moles.

c) Because CO_2 is a gas, we will write $[CO_2]_{(g)}$ as P_{CO_2}, i.e. the partial pressure of CO_2 gas in the atmosphere. If we assume that P_{CO_2} = 1 bar, we can calculate the value of K_1:

$$K_1 = \frac{[H_2CO_3]_{(aq)}}{P_{CO_2}} = \frac{\text{moles}(H_2CO_3)}{1} = \frac{0.0336}{1} = 10^{-1.47}$$

For the last step, convert the value to a power of 10 (i.e. give the answer as 10^X) by taking the log of the number. (K is a ratio and has no units.)

3. Now we will **calculate the pH** of water in which H_2CO_3 is dissociating, according to Reaction 2. (See Misra, Ch. 7.5.1)

a) First we will determine the value of $[H_2CO_3]_{(aq)}$, at the atmospheric pressure of CO_2 , $P_{CO_2} = 10^{-3.5}$ bar.

Use the expression for K_1 determined at the top of the page to calculate $[H_2CO_3]_{(aq)}$:

$$[H_2CO_3]_{(aq)} = K_1 * P_{CO_2} = 10^{-1.47} * 10^{-3.5} = 10^{-4.97}$$

(Keep everything in powers of 10. Use the additive property of powers, i.e. $10^X * 10^Y = 10^{(X+Y)}$)

b) Rearrange the expression for K_2 that you wrote in Q1 above, so that we can determine the concentration of H^+ ions:

$$[H^+] = \frac{K_2 * [H_2CO_3]_{(aq)}}{[HCO_3^-]}$$

c) Without going into details (see Misra for an explanation) we will assume that $[HCO_3^-] = [H^+]$.

Combine this with the equation above (b), so that it is now written in terms of $[H^+]$, $[H_2CO_3]_{(aq)}$ and K_2.

$$[H^+] = \frac{K_2 * [H_2CO_3]_{(aq)}}{[H^+]} \qquad [H^+]^2 = K_2 * [H_2CO_3]_{(aq)}$$

d) Given that $K_2 = 10^{-6.35}$, calculate $[H^+]$:
(Again, keep everything in powers of 10. Again use the additive property of powers, $\sqrt{10^X} = 10^{X/2}$)

$$[H^+]^2 = K_2 * [H_2CO_3]_{(aq)} = 10^{-6.35} * 10^{-4.97} = 10^{-11.32}$$

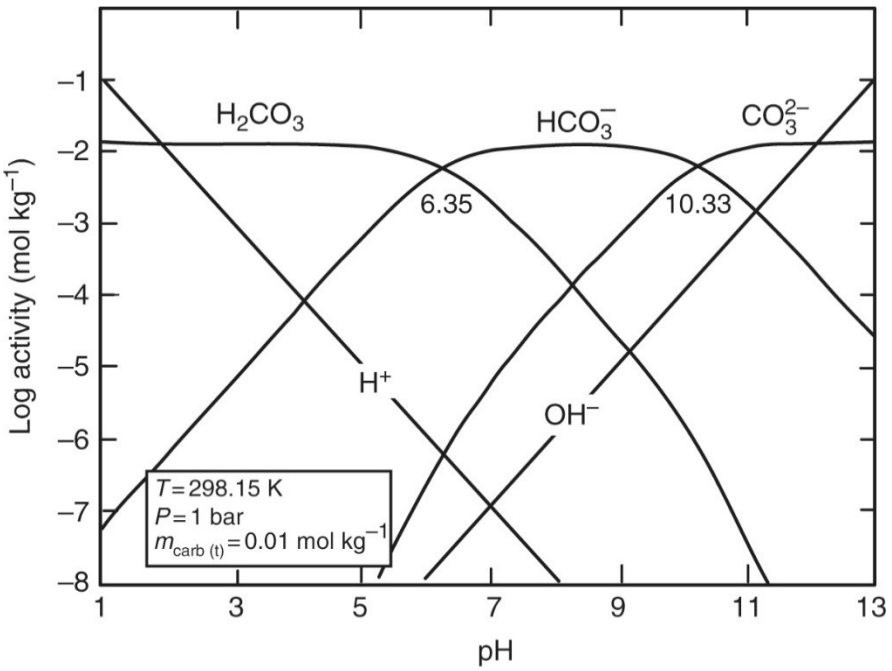
$$[H^+] = 10^{-5.66}$$

e) Convert $[H^+]$ into pH using the relationship, $pH = -\log_{10}[H^+]$

pH = 5.66

How would you describe this solution? strongly / mildly acidic / neutral / basic

f) Reactions 1-3, in combination with the water dissociation reaction, can be used to calculate the dominant ion species in a CO_2-H_2O solution as a function of pH. This information is plotted in the figure below. Total dissolved carbonate (m_{carb}) = 0.01 mol kg^{-1}.



Examine the graph and answer these questions:

i) Do the concentrations of H^+ and OH^- ions do what you would expect as a function of pH? – explain.

Hopefully they do what you expect! H^+ increases with decreasing pH, and OH^- increases with increasing pH. This is what you would expect since pH = -log $[H^+]$. The H^+ line is a straight line of slope -1 because the y axis is essentially showing log $[H^+]$.

ii) Explain how the graph shows that a solution is "neutral" at pH = 7.

At pH 7, the amounts of H^+ and OH^- ions are equal. So the solution is neither acidic nor basic. At pH 7, log $[H^+]$ = log $[OH^-]$ = -7.

iii) Above (e) you calculated the pH of water in equilibrium with the atmosphere. Show where this lies on the diagram. What is the dominant (highest concentration) species at this pH?

From above, pH = 5.66. The dominant species is H_2CO_3 (aq).

iv) The pH of the oceans is about 8. Show where this lies on the diagram. What is the dominant (highest concentration) species at this pH?

For pH = 8, The dominant species is HCO_3^- (bicarbonate ions).

How would you describe the ocean: stongly / mildly acidic / neutral / basic.

v) In what kind of solution (acidic, basic, etc) would you find a high concentration of carbonate ions (CO_3^{2-})? - In a basic solution, e.g. pH >11.

B. Weathering Reactions

1. Dissolution Reactions. In a dissolution reaction, a mineral simply dissolves into water. Reactions are usually driven by acidic water. Carbonate rocks are very susceptible to chemical weathering of this type. Carbonic acid (see Part A) is an important acid that drives the reaction.

For calcite dissolution, the reaction is:

$$CaCO_{3\,(s)} + H_2CO_{3\,(aq)} = Ca^{2+} + 2HCO_3^-$$

This reaction is balanced. Notice that the CHARGES must balance, as well as the elements.

For dolomite dissolution, the reaction is:

$$CaMg(CO_3)_{2\,(s)} + 2\ H_2CO_{3\,(aq)} = Ca^{2+} + Mg^{2+} + 4\ HCO_3^-$$

This reaction is NOT balanced – your task is to balance it.

Why do rivers and groundwaters contain significant amounts of Ca^{2+} and HCO_3^- ions? How would concentrations of these ions change if there was more CO_2 in the atmosphere?

CO_2 in the atmosphere dissolves in water, forming carbonic acid, H_2CO_3 (Reaction 1 above). When carbonic acid dissociates, it forms bicarbonate ions (HCO_3^-) and H^+ ions (Reaction 2 above). H+ ions mean that the solution is acidic, and this enhances the dissolution of carbonate minerals. When carbonate minerals dissolve in water, the water becomes rich in Ca^{2+} and HCO_3^- ions.

If there was more CO_2 in the atmosphere, water would be more acidic and this would increase the dissolution rate of carbonate minerals, thus increasing the amounts of these dissolved ions in water.

2. Hydrolysis Reactions. Most silicate minerals weather in reactions that have one or more solid reaction products, in addition to dissolved species. A typical reaction is weathering of potassium feldspar into the clay mineral, kaolinite. This reaction occurs in an acidic solution. The reaction products include K^+ ions in solution, and silicic acid, H_4SiO_4. (Silicic acid is a weak acid and behaves in a similar way to carbonic acid – see below.)

The reaction is:

$$2\ KAlSi_3O_{8\,(s)} + 9\ H_2O_{(l)} + 2\ H^+ = Al_2Si_2O_5(OH)_{4\,(s)} + 2\ K^+ + 4\ H_4SiO_{4\,(aq)}$$

K-feldspar + water + acid = kaolinite + potassium ions + silicic acid

This reaction is NOT balanced – your task is to balance it.

If you are having trouble with this, try balancing the elements in the following order:

Al, Si, O, H, K

3. Hydration Reactions and Solubility. These reactions convert an anhydrous (water-free) mineral into a hydrated mineral. An example is seen in calcium sulphate minerals, in the reaction of anhydrite ($CaSO_4$) to gypsum ($CaSO_4.2H_2O$). Anhydrite is formed in evaporite deposits, along with halite (NaCl).

a) Formation of an evaporite occurs when water evaporates from a lake. The water becomes saturated in salt ions, and salts precipitate.

For NaCl, $Na^+_{(aq)} + Cl^-_{(aq)} = NaCl_{(s)}$

Write the reaction for formation of anhydrite:

$Ca^{2+} + SO_4^{2-} = CaSO_4$

Now write the reaction that describes hydration of anhydrite to form gypsum:

$CaSO_4 + 2H_2O = CaSO_4.2H_2O$

(This transformation is reversible, with gypsum forming anhydrite by heating to around 200 °C under normal atmospheric conditions.)

b) Both halite and anhydrite are soluble in water.

For NaCl, the solubility product K_{sp} is given by: $K_{sp} = [Na^+] * [Cl^-]$

Write the equation for the solubility product of anhydrite:

$K_{sp} = [Ca^{2+}] * [SO_4^{2-}]$

The equation for the solubility product of gypsum is the same as anhydrite, because the water in gypsum just becomes part of the solvent.

The solubility product for halite at 25 °C has a value of 36 ($K_{sp} = 36$).

This means that for solid halite in equilibrium with water, the concentration of Na (in mol L^{-1}) times the concentration of Cl (in mol L^{-1}) equals 36.

$K_{sp} = [Na^+] [Cl^-] = 36$ so $[Na^+] = [Cl^-] = 6$ mol L^{-1}

In other words, a solution saturated in NaCl contains 6 mol L^{-1} Na+ ions and 6 mol L^{-1} Cl ions.
This is equivalent to 6 moles of NaCl.

Given that the atomic masses are Na = 23 and Cl = 35.5, calculate the total mass of NaCl that can be dissolved in 1 litre of water: (no. of moles x molecular mass)

Molecular mass = 23 + 35.5 = 58.5.

6 moles = 58.5 * 6 = 351 g. (units?)

The solubility product for gypsum at 25 °C is 2.4×10^{-5}. What mass of gypsum will dissolve in 1 litre of water? (Z for Ca = 40, S = 32, O = 16)

$\sqrt{(2.4 * 10^{-5})} = 0.005$ moles of gypsum. Molecular mass of gypsum = 40 + 32 + 64 + 2(18) = 172.

172 * 0.005 = 0.86 g